

期望值是博弈的产物，而非游戏规则固有的属性。

definition 反身性，即 chernoff 策略被破，并且导致所有参与者使用随机数，这导致将毁灭，游戏回到最初的“平等”状态。

在由具有将理性行为的群体构成的系统(市场)中，理性和随机性本身可以成为一种发展相对优势的策略。

18.4. Great Expectations

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The payoffs for each outcome are the same in Figures 18.6 and 18.7, but the probabilities of the outcomes are different. For example, it is no longer possible for all three players to guess correctly, since Nick and Eric are always guessing differently. More importantly, the outcome where your payoff is \$4 is also no longer possible. Since Nick and Eric are always guessing differently, one of them will always get a share of the pot. As you might imagine, this is not good for you!

When we use equation (18.3) to compute your expected return in the collusion scenario, we find that

$$\begin{aligned}\text{Ex}[\text{payoff}] &= 0 \cdot 0 + 1 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} + 4 \cdot 0 \\ &\quad + (-2) \cdot 0 + (-2) \cdot \frac{1}{4} + (-2) \cdot \frac{1}{4} + 0 \cdot 0 \\ &= -\frac{1}{2}.\end{aligned}$$

So watch out for these biker dudes! By colluding, Nick and Eric have made it so that you expect to lose \$0.50 every time you play. No wonder you lost \$500 over the course of 1000 wagers.

How to Win the Lottery

Similar opportunities to collude arise in many betting games. For example, consider the typical weekly football betting pool, where each participant wagers \$10 and the participants that pick the most games correctly split a large pot. The pool seems fair if you think of it as in Figure 18.6. But, in fact, if two or more players collude by guessing differently, they can get an "unfair" advantage at your expense!

In some cases, the collusion is inadvertent and you can profit from it. For example, many years ago, a former MIT Professor of Mathematics named Herman Chernoff figured out a way to make money by playing the state lottery. This was surprising since the state usually takes a large share of the wagers before paying the winners, and so the expected return from a lottery ticket is typically pretty poor. So how did Chernoff find a way to make money? It turned out to be easy!

In a typical state lottery,

- all players pay \$1 to play and select 4 numbers from 1 to 36,
- the state draws 4 numbers from 1 to 36 uniformly at random,
- the state divides 1/2 of the money collected among the people who guessed correctly and spends the other half redecorating the governor's residence.

This is a lot like the game you played with Nick and Eric, except that there are more players and more choices. Chernoff discovered that a small set of numbers